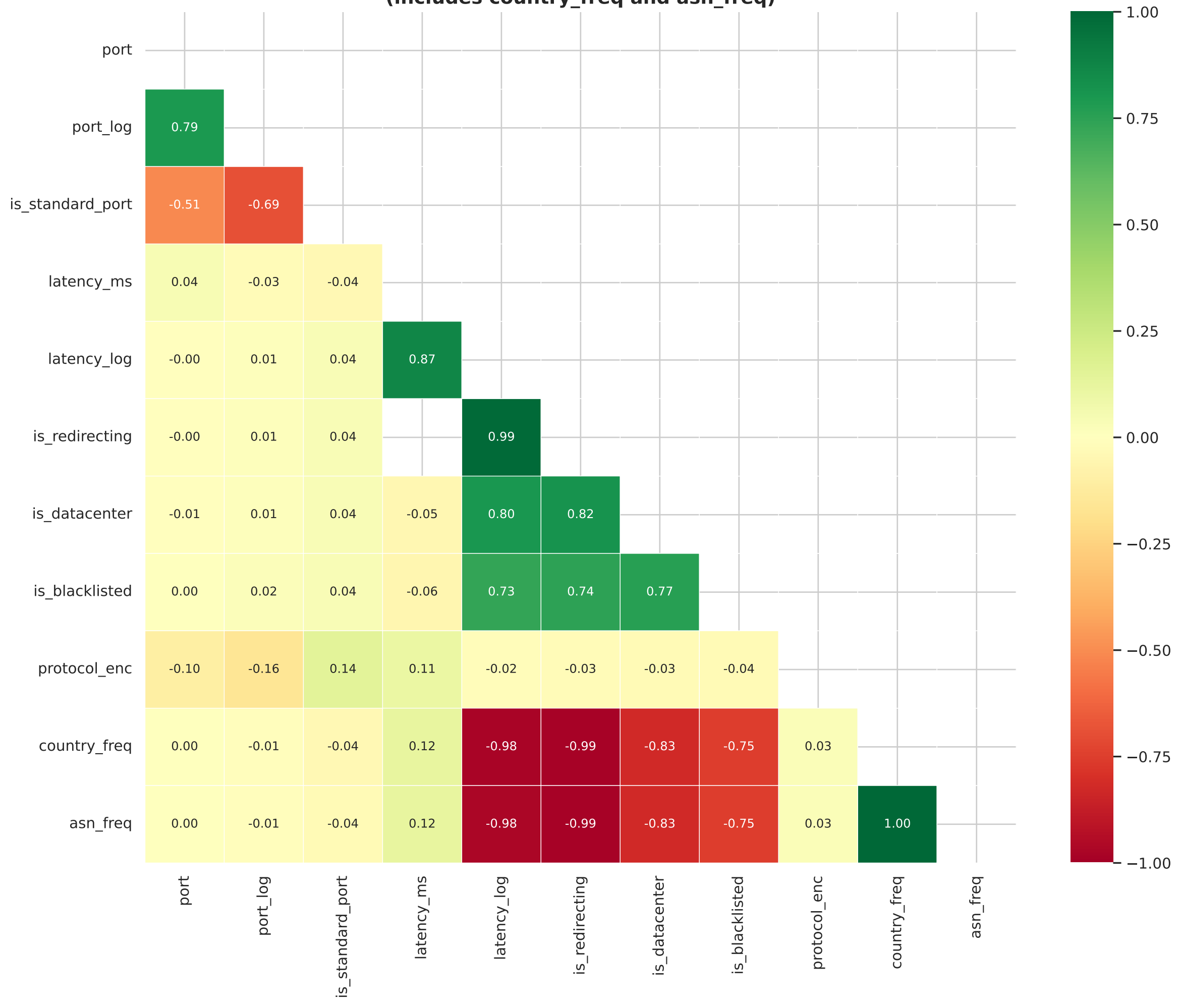


1. Dataset Overview & Feature Engineering

```
Dataset shape      : (115042, 35)
Working configs    : 1,451 / 115,042
Datacenter exits   : 973
Unique ASNs        : 197
```

```
Security check coverage (working configs only):
  proxy_detected   : 814 / 1451
  has_ipv6         : 393 / 1451
  ipv6_leak        : 57 / 1451
```

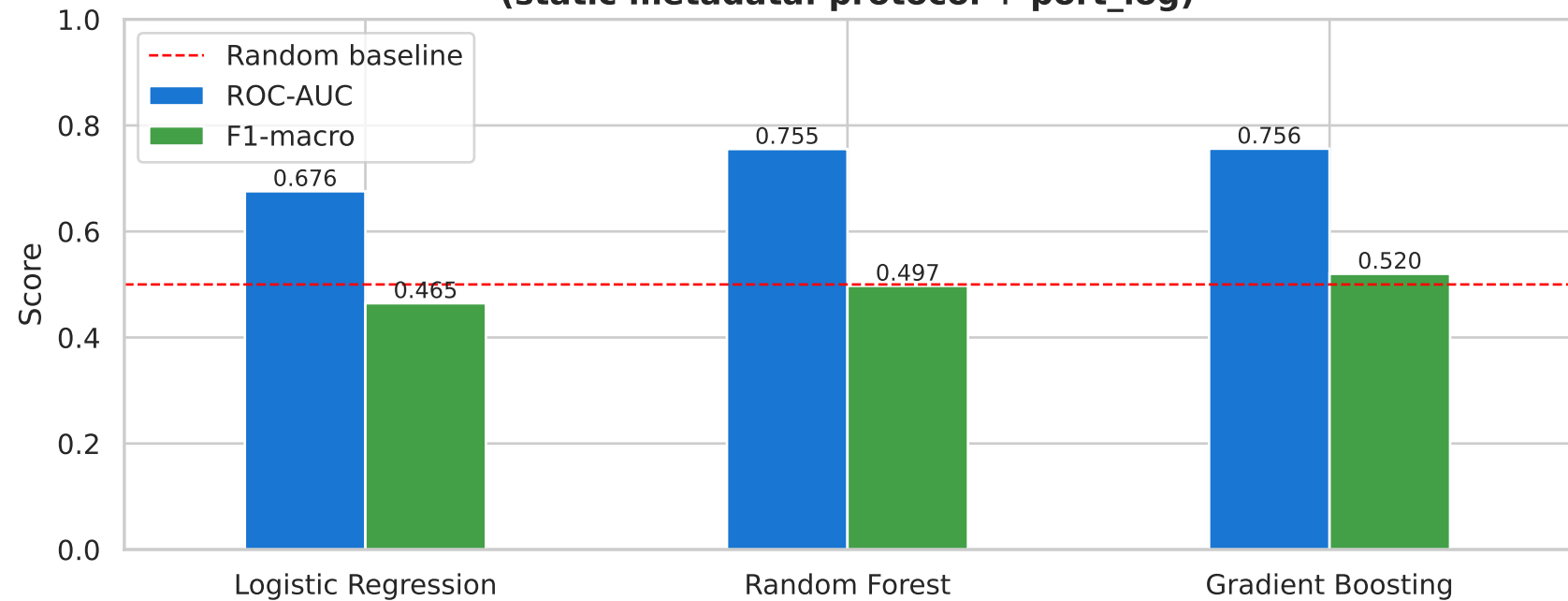
Feature Correlation Heatmap
(includes country_freq and asn_freq)



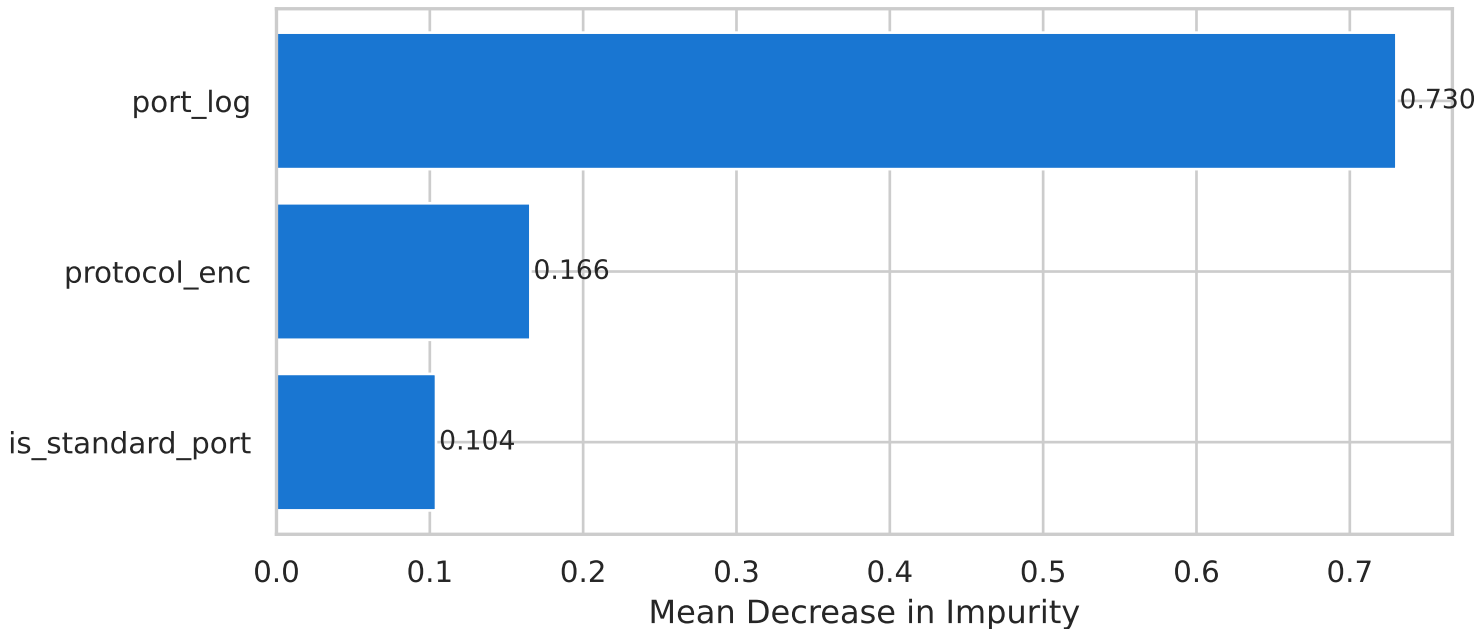
3. Availability Prediction — 5-Fold CV Scores

	ROC-AUC	F1-macro
Logistic Regression	0.675756	0.464540
Random Forest	0.755445	0.497343
Gradient Boosting	0.756069	0.520028

Availability Prediction — 5-Fold CV (static metadata: protocol + port_log)



Feature Importance — Predicting Availability (Random Forest, static metadata)

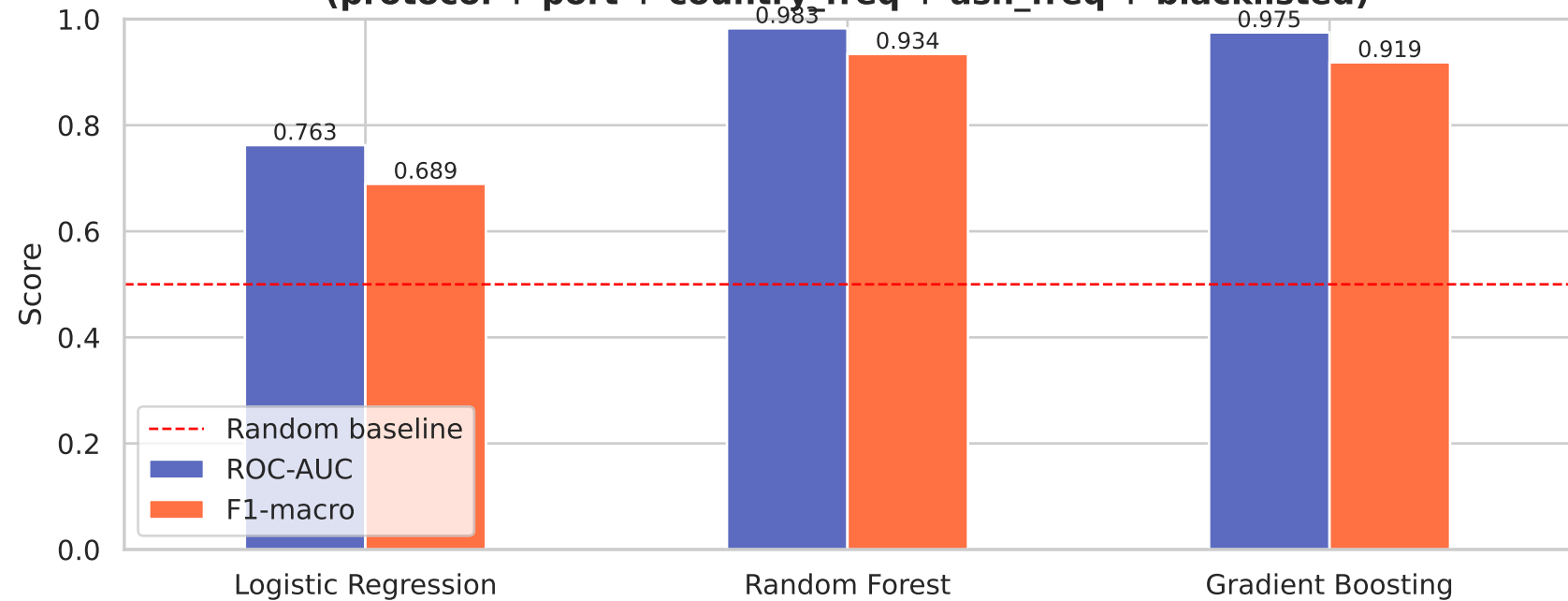


5. Datacenter Exit Prediction — CV Scores

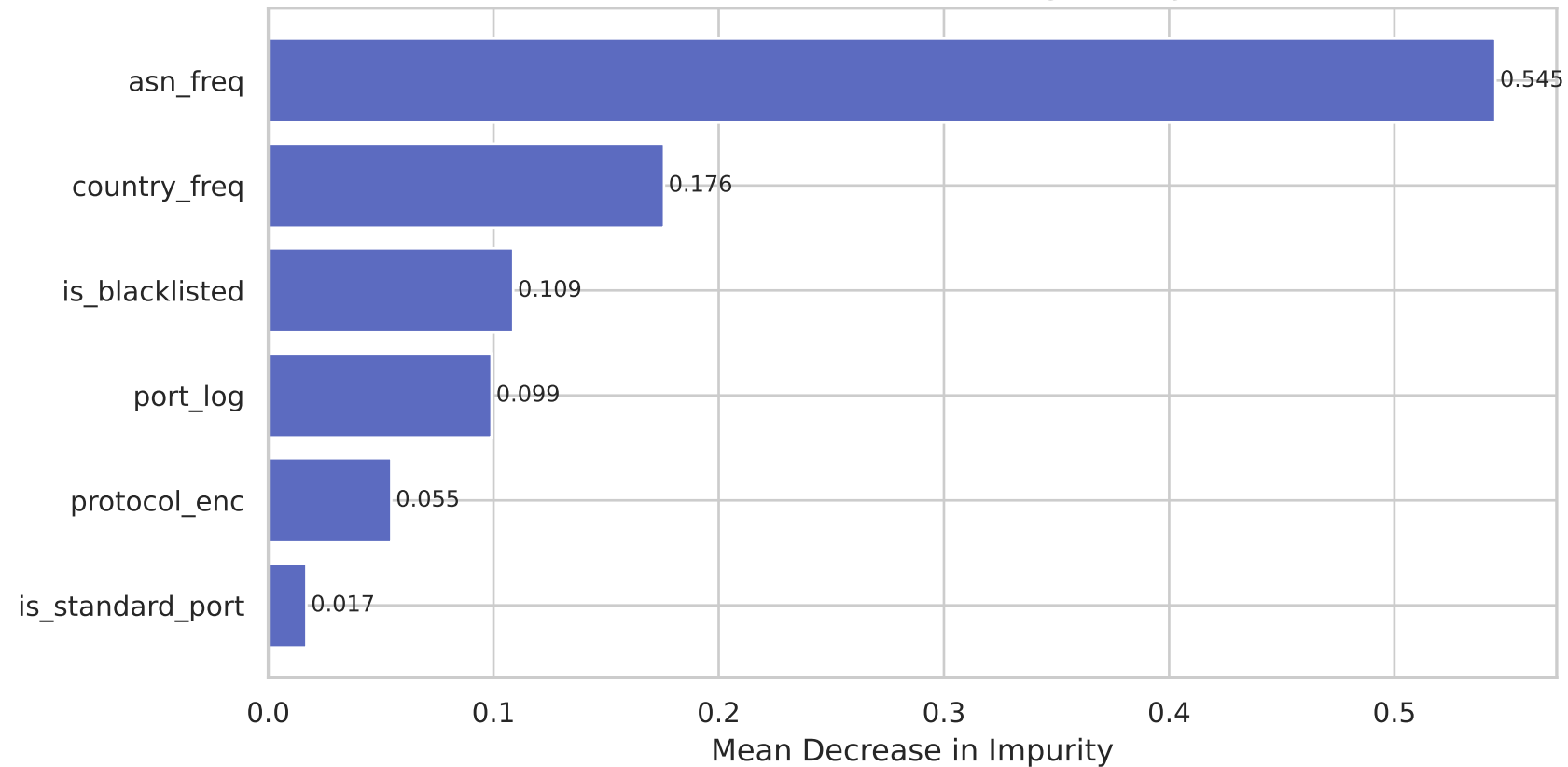
Dataset: 1451 configs | datacenter=973 residential=478

	ROC-AUC	F1-macro
Logistic Regression	0.762519	0.689438
Random Forest	0.982661	0.934426
Gradient Boosting	0.974902	0.918558

Datacenter Exit Prediction — 5-Fold CV
(protocol + port + country_freq + asn_freq + blacklisted)



Feature Importance — Predicting Datacenter Exit (Random Forest, working configs)



7. Latency Regression — Results

Latency Regression (log-scale, target-encoded country):

$$R^2 = -0.032 \pm 0.098$$

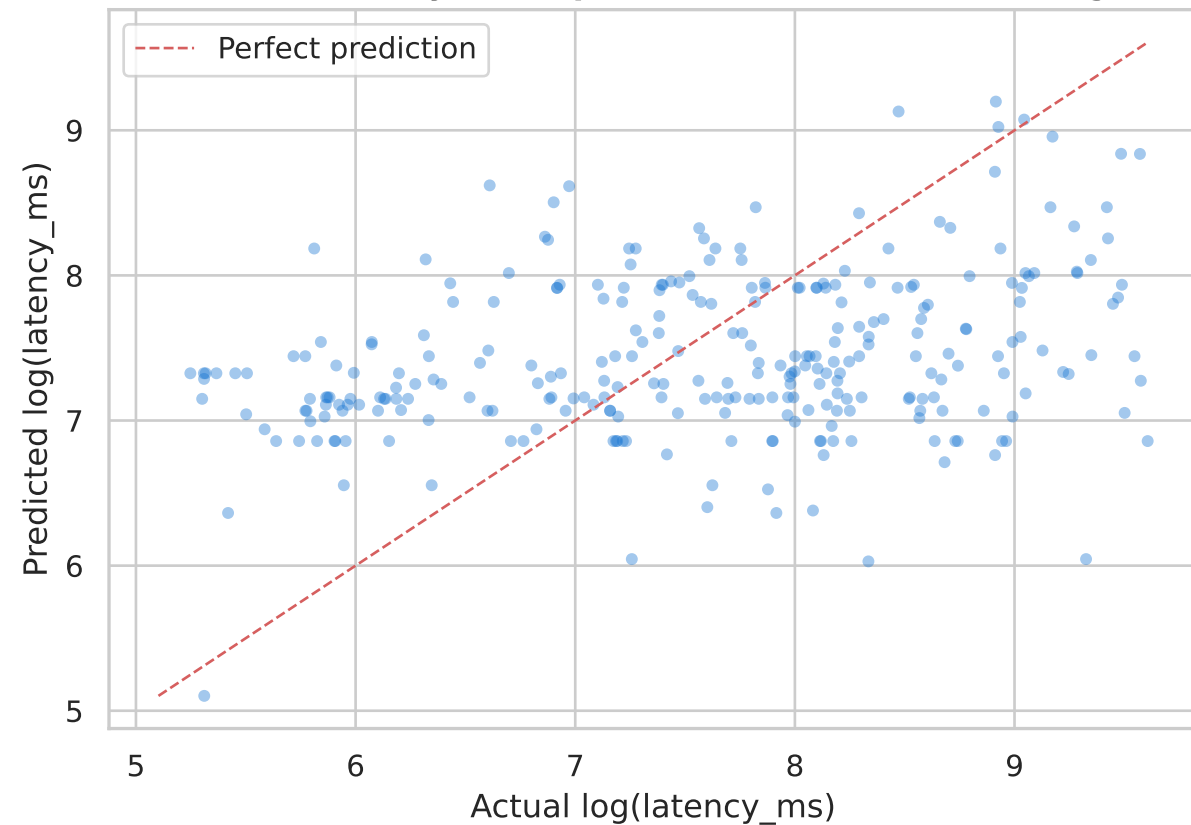
$$\text{MAE} = 0.915 \pm 0.058 \quad (\text{log-ms units})$$

$R^2 < 0$ means the model is worse than predicting the mean.

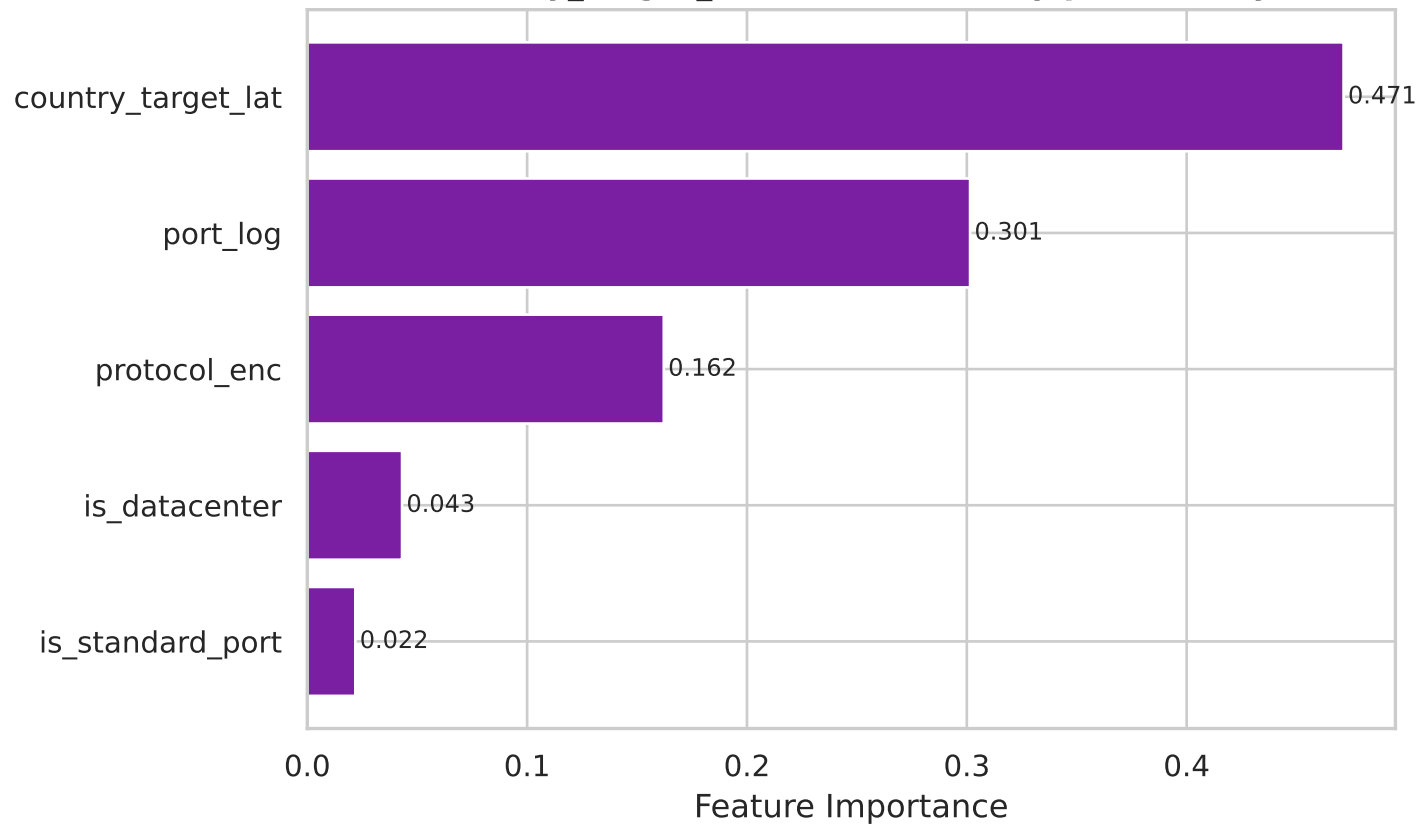
Latency is driven by real-time network conditions (server load, routing path, congestion) that are invisible in static config metadata. No amount of ML will fix this; the signal is not there.

Latency Regression — Test Set

CV $R^2=-0.032$ — latency is not predictable from static config metadata



Latency Feature Importance (country_target_lat = mean latency per country)



8. K-Means Silhouette Scores by k

Silhouette scores (higher = better separated clusters):

k= 2 0.6731 ← best

k= 3 0.3040

k= 4 0.2836

k= 5 0.3007

k= 6 0.3157

k= 7 0.3423

k= 8 0.3505

k= 9 0.3562

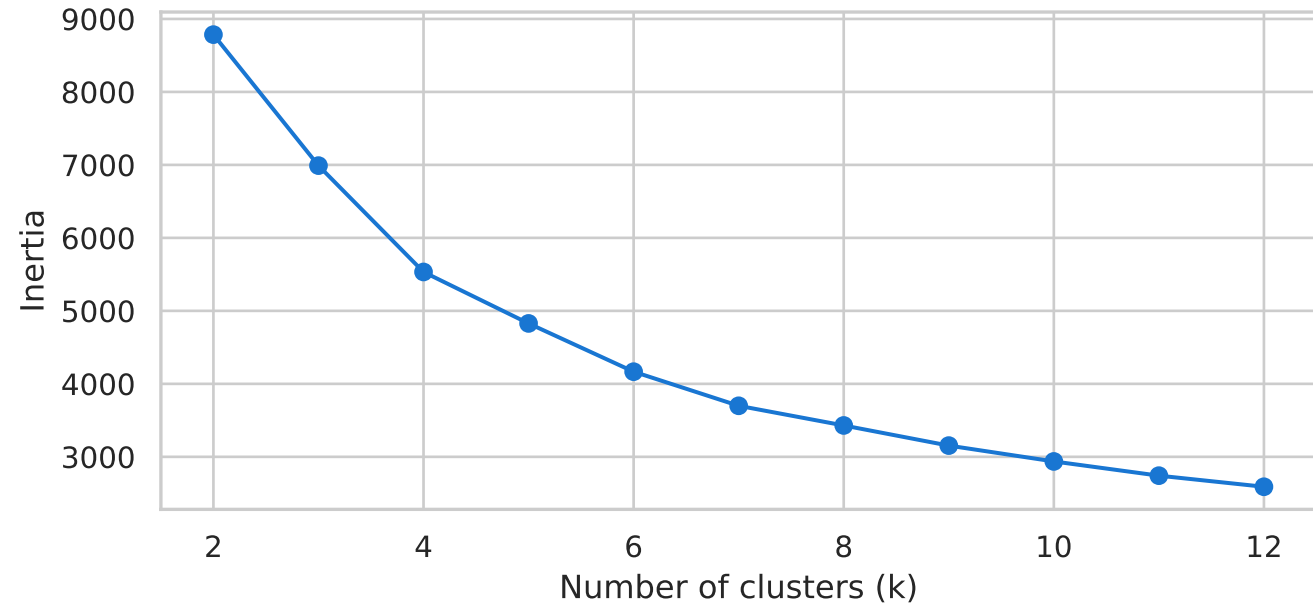
k=10 0.3538

k=11 0.3509

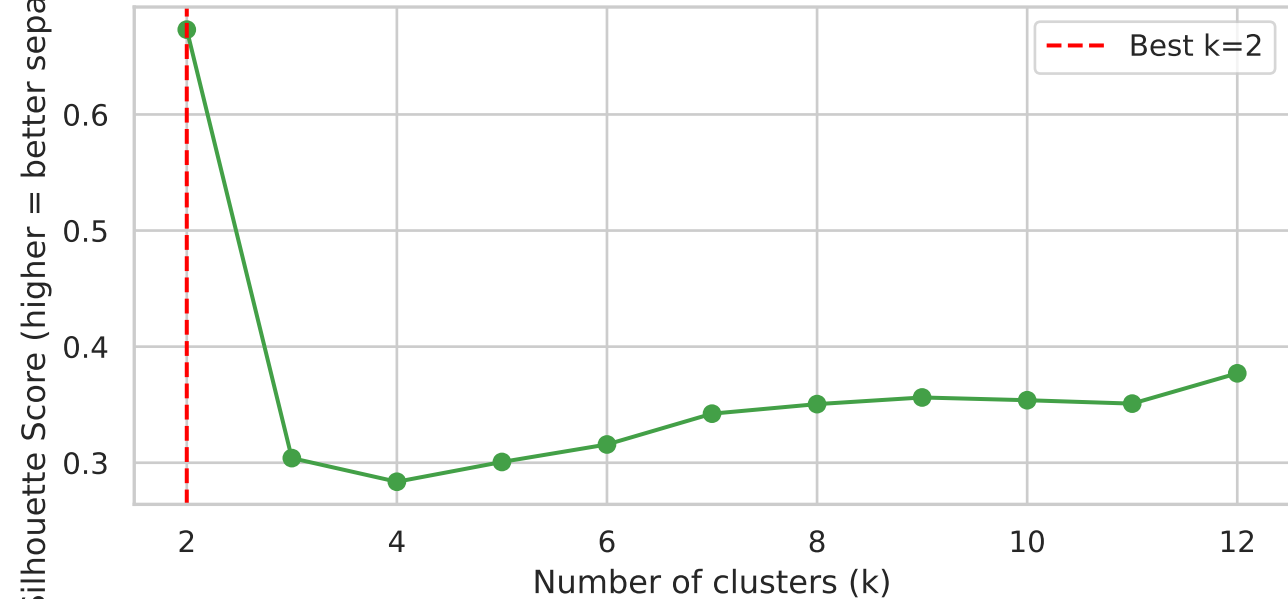
k=12 0.3770

K-Means — Choosing Optimal k

Elbow Curve



Silhouette Score — picks optimal k objectively



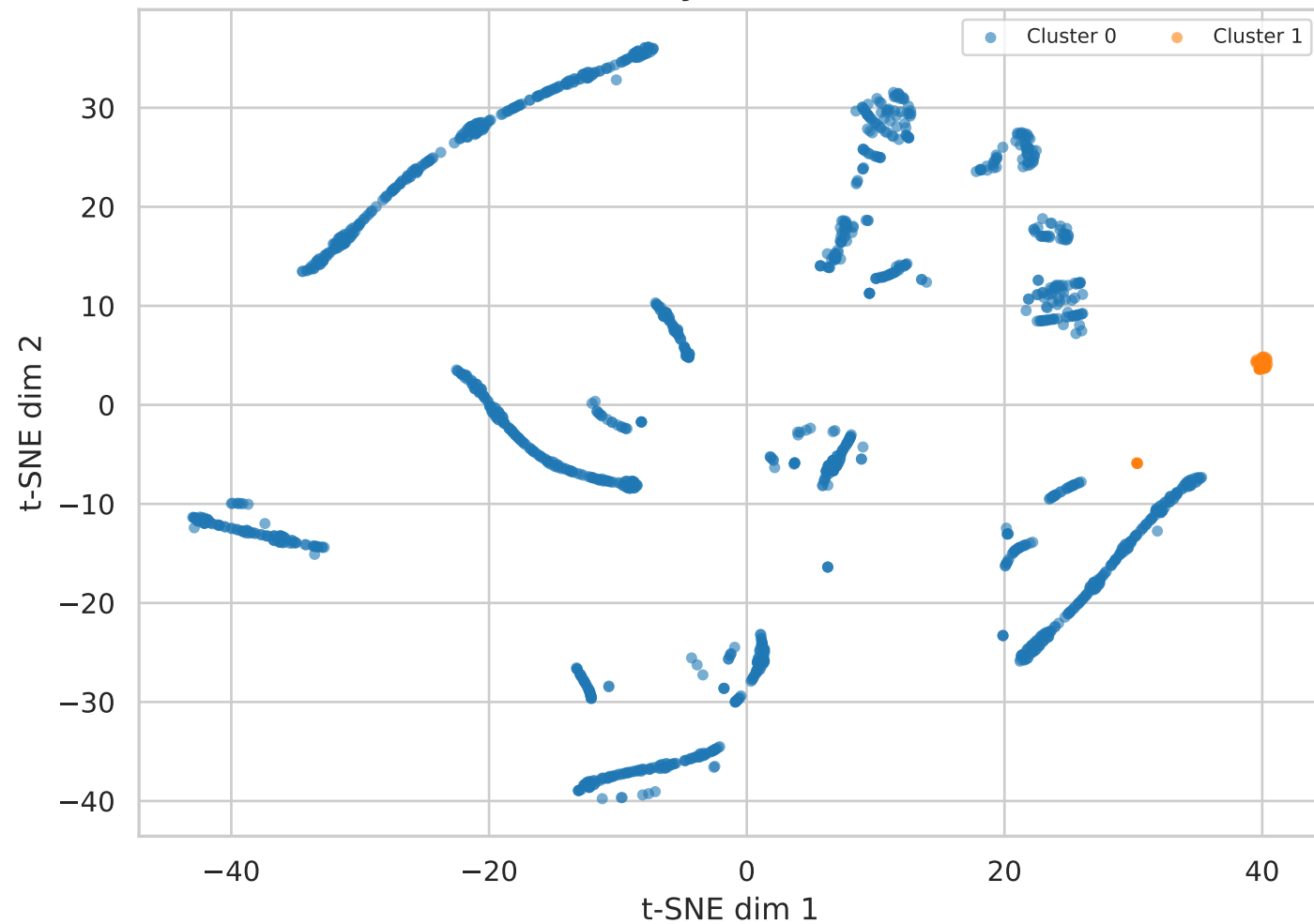
8b. Cluster Profile (k=2)

k=2 selected by silhouette score

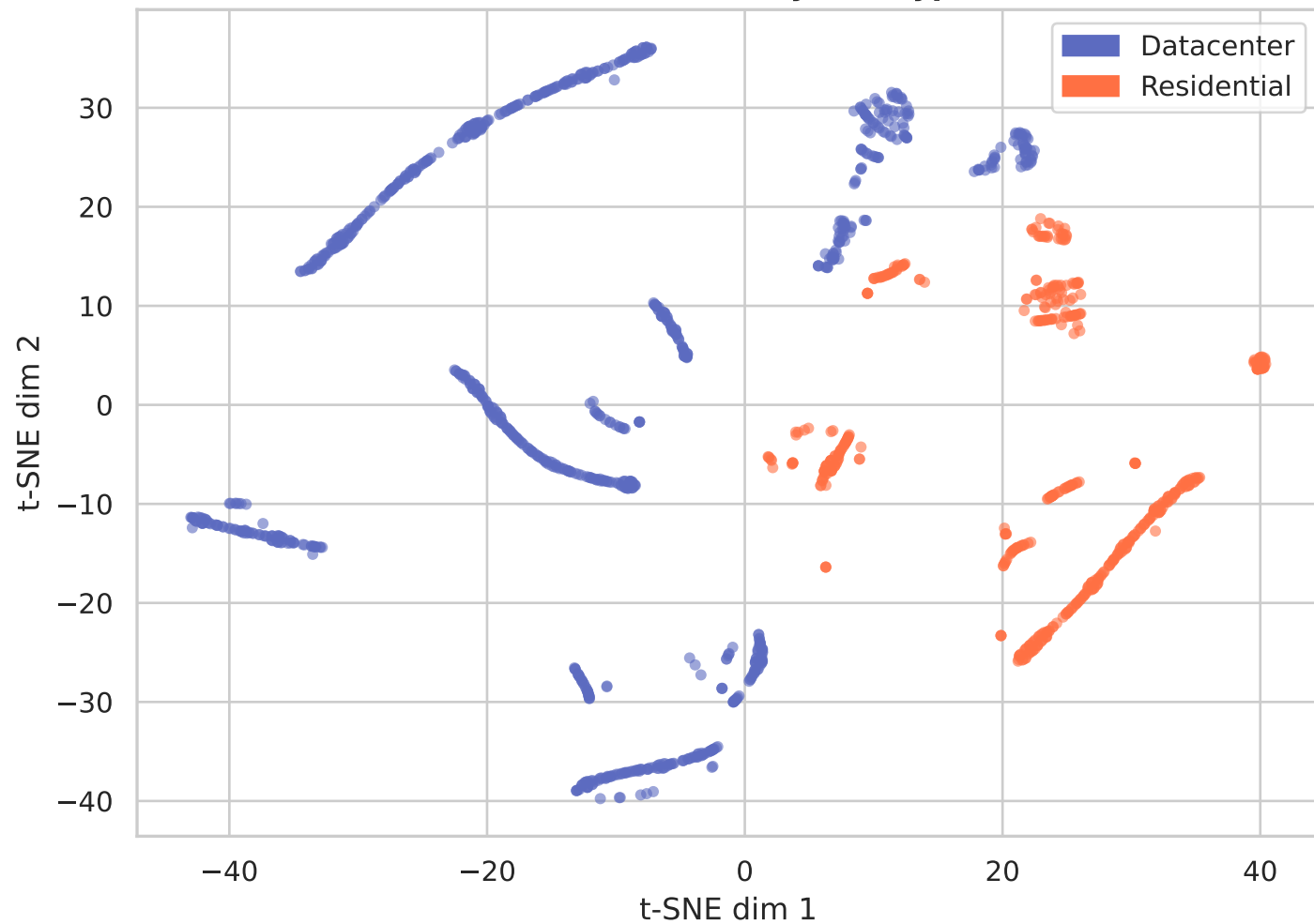
	count	avg_latency	pct_datacenter	pct_blacklist	top_protocol	top_country	pct_proxy_det	pct_ipv6
cluster								
0	1418	2966.512	68.6	57.0	vless	Germany	56.8	27.4
1	33	5446.651	0.0	0.0	vless	United Kingdom	27.3	15.2

Config Space Visualization (t-SNE, working configs only)

t-SNE — Colored by K-Means Cluster (k=2)

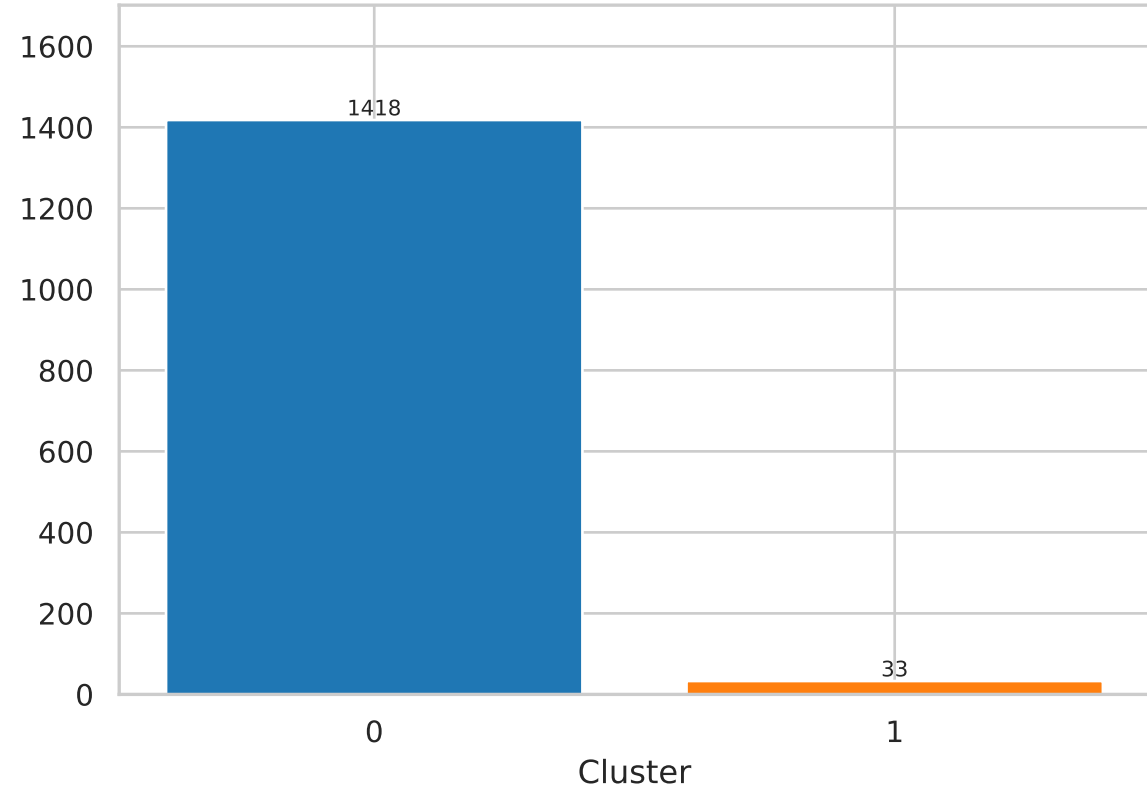


t-SNE — Colored by Exit Type

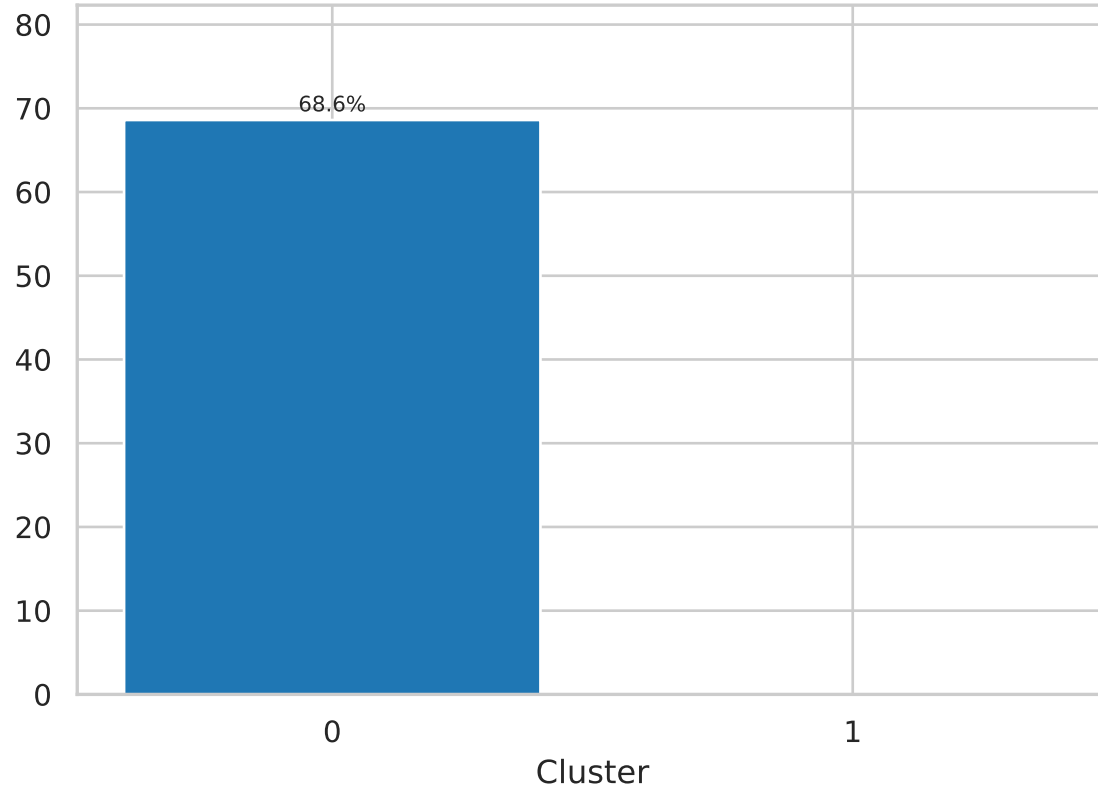


Cluster Profiles — Key Metrics

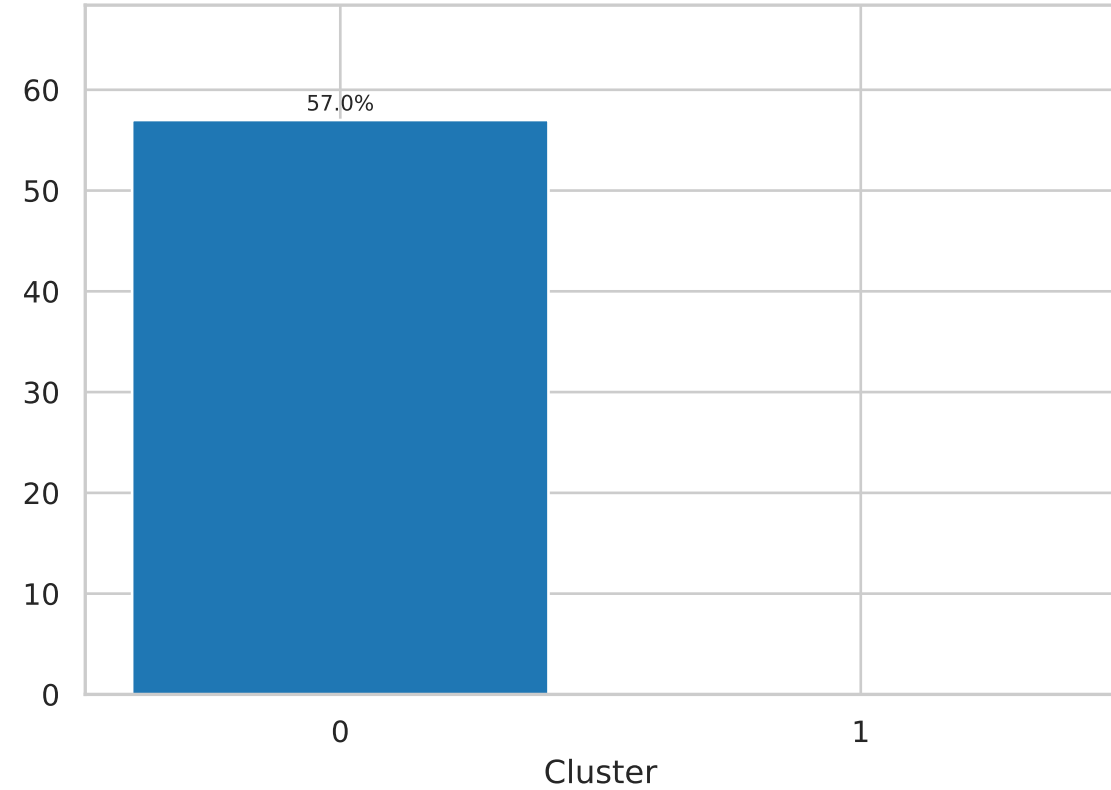
Config Count



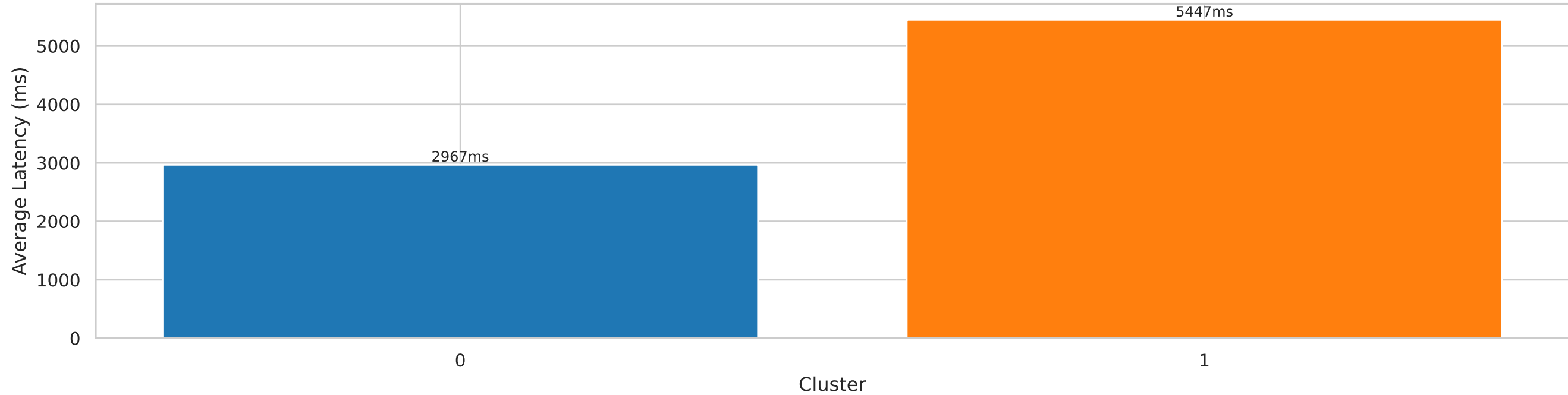
Datacenter Exit (%)



Blacklisted IP (%)



Average Latency per Cluster



11. Anomaly Detection — Breakdown

Normal configs : 1380

Anomalous configs: 71

Anomaly breakdown:

Protocol distribution:

protocol

vless 33

ss 14

hysteria2 13

vmess 10

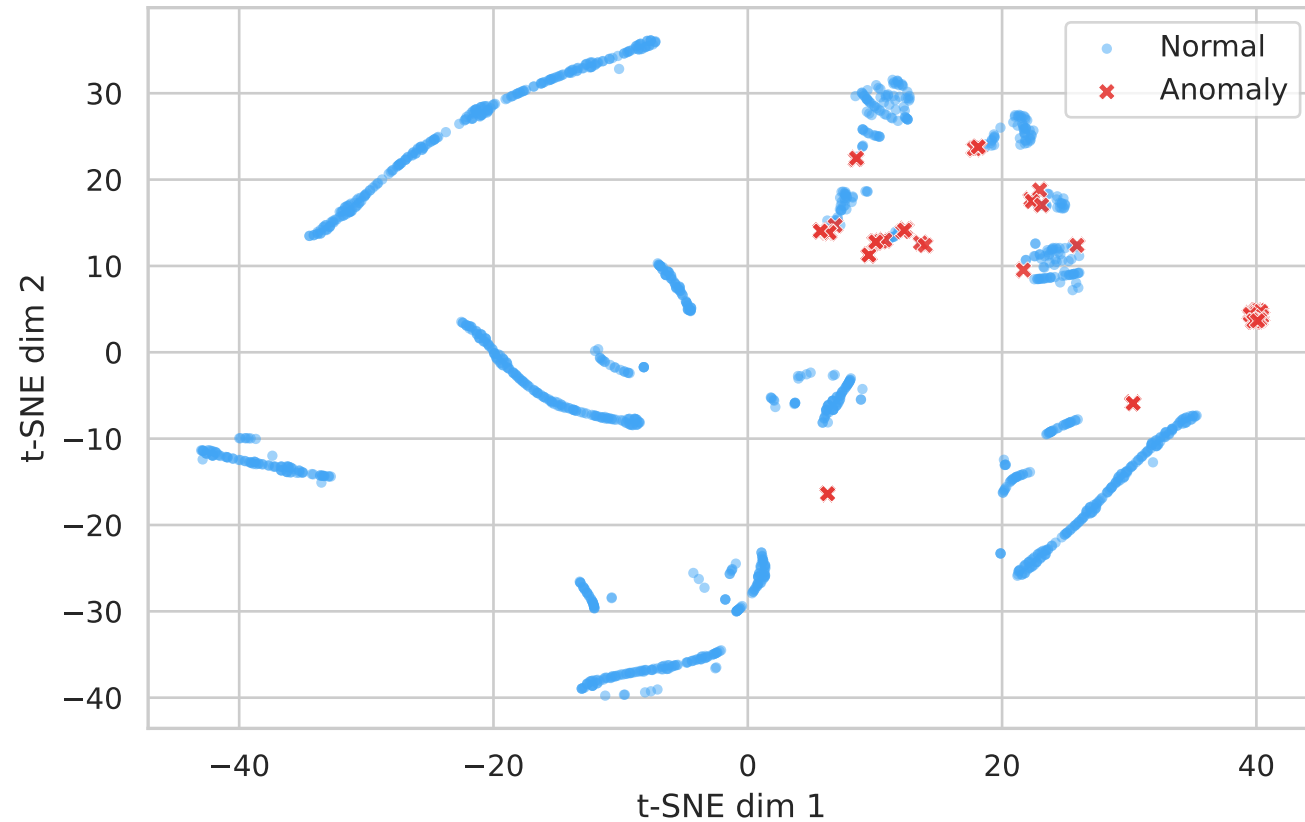
trojan 1

Avg latency : 3552ms (normal: 2996ms)

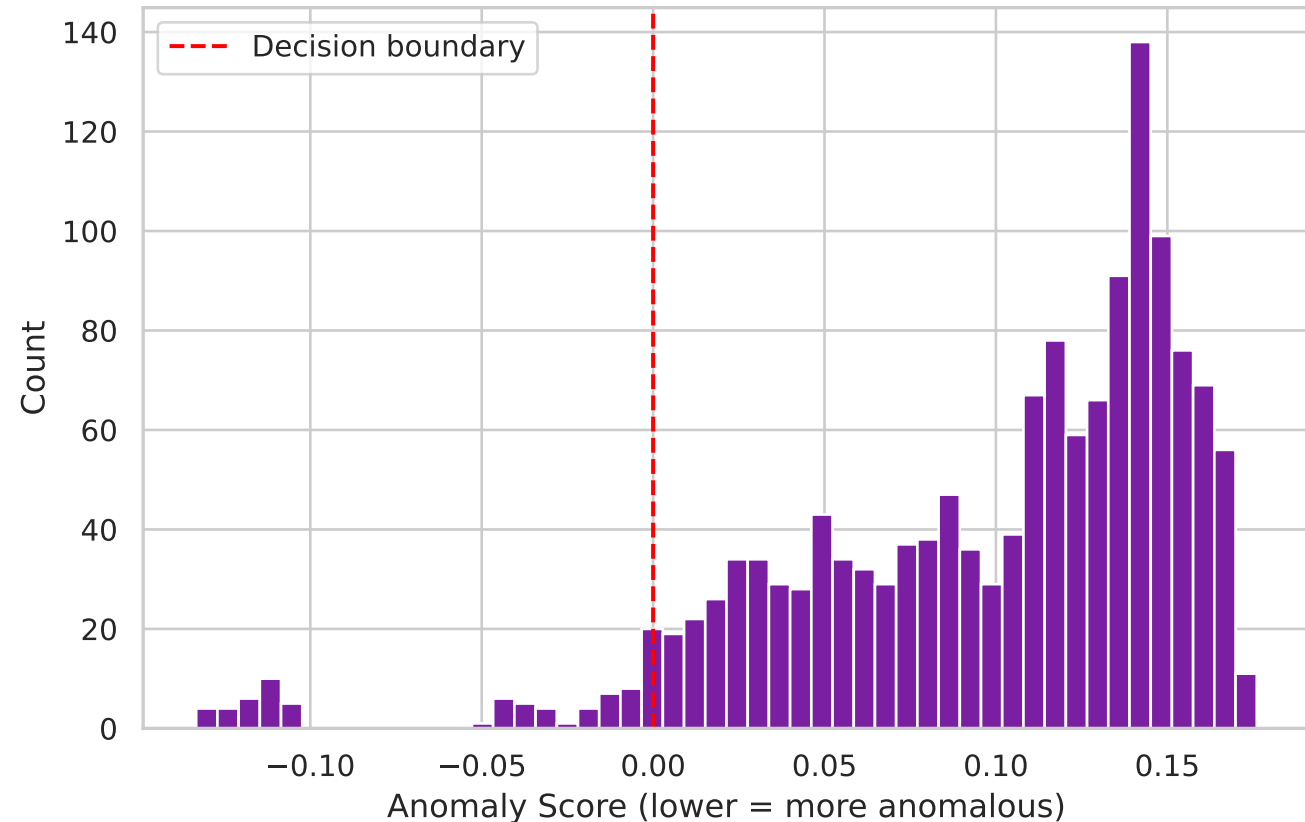
Datacenter pct : 21.1% (normal: 69.4%)

Blacklisted pct: 36.6% (normal: 56.7%)

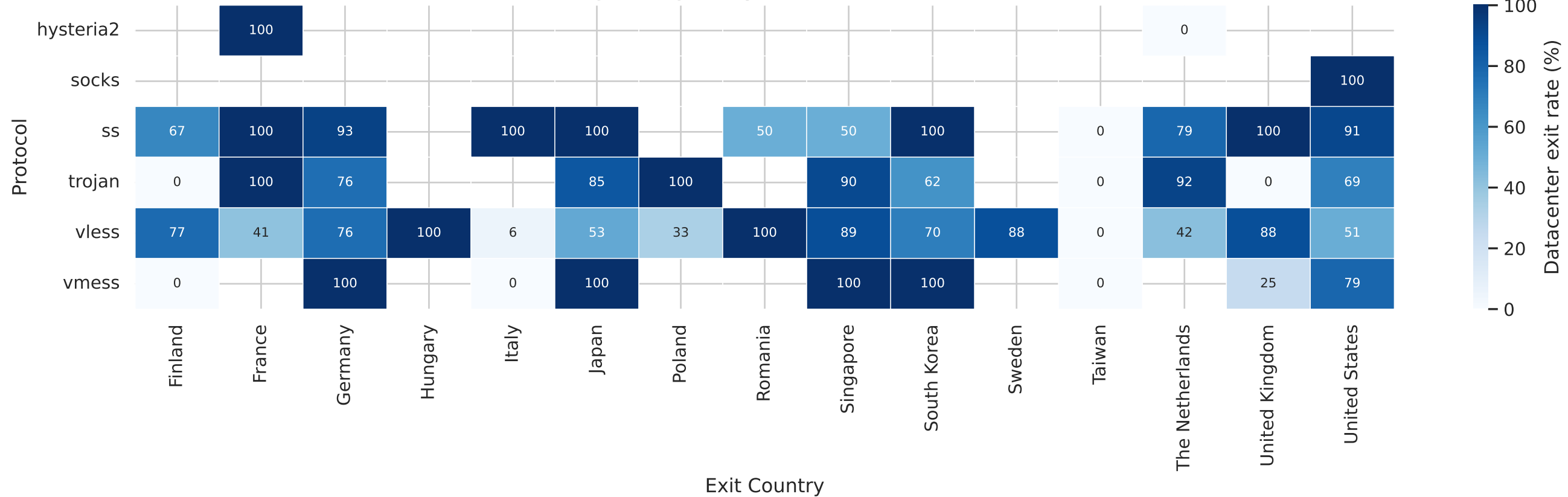
**Anomaly Detection (Isolation Forest)
t-SNE projection**



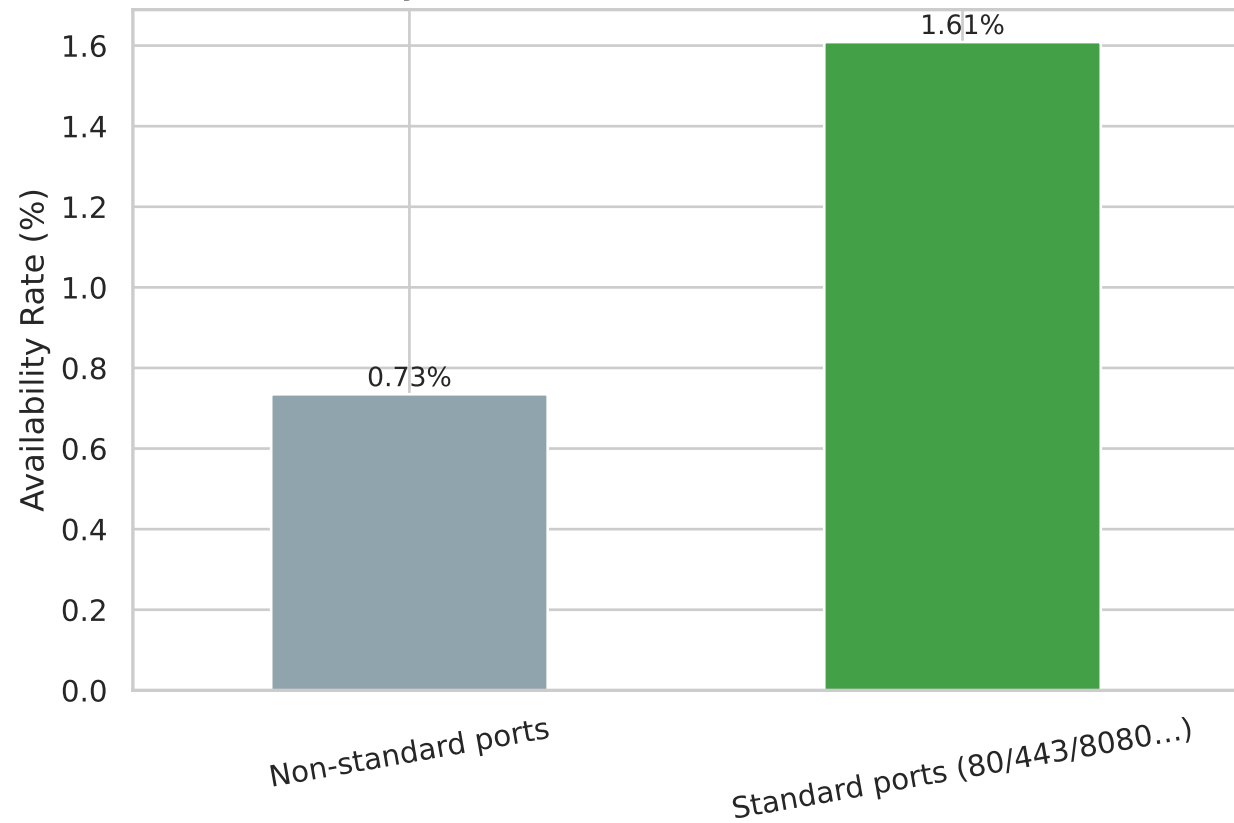
Anomaly Score Distribution



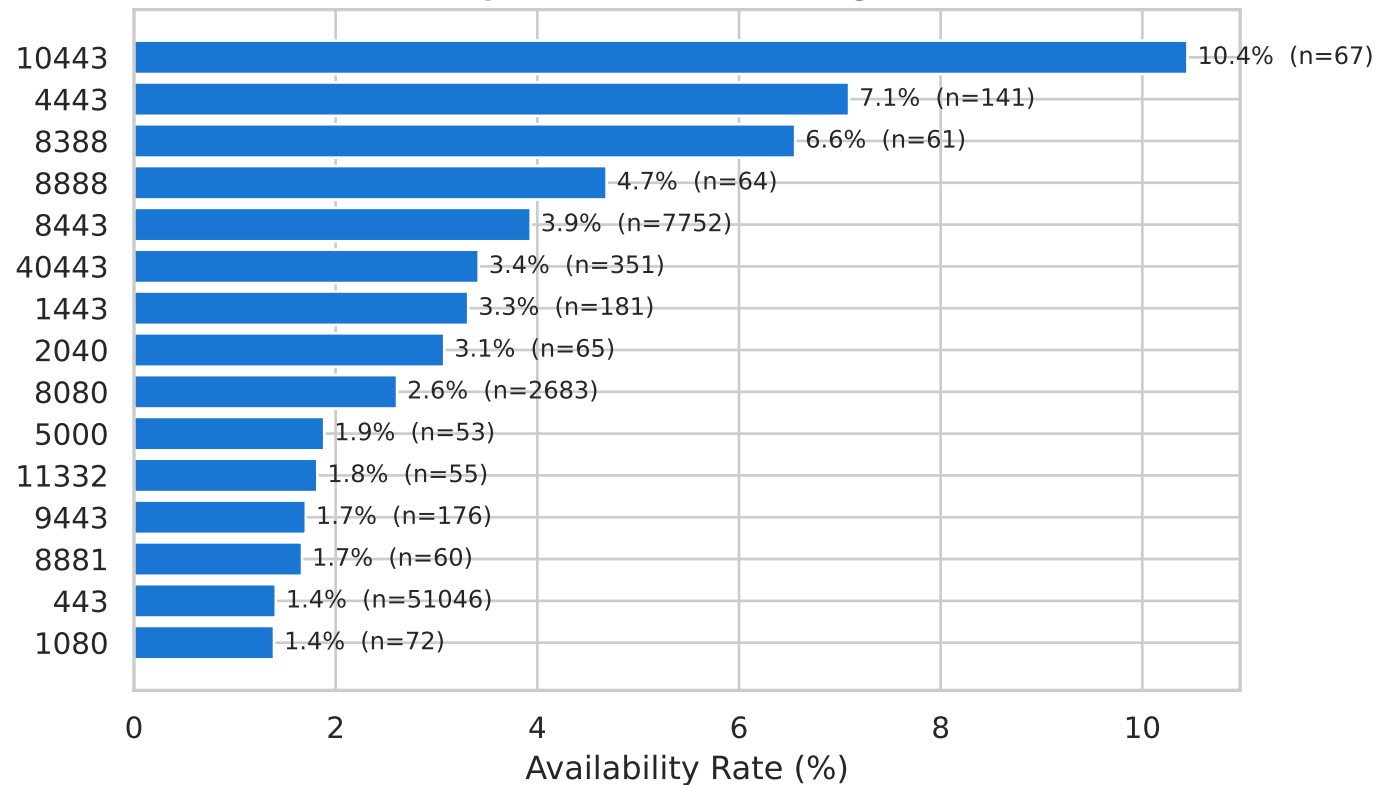
Datacenter Exit Rate (%) — Protocol × Country
(working configs only, top 15 exit countries)



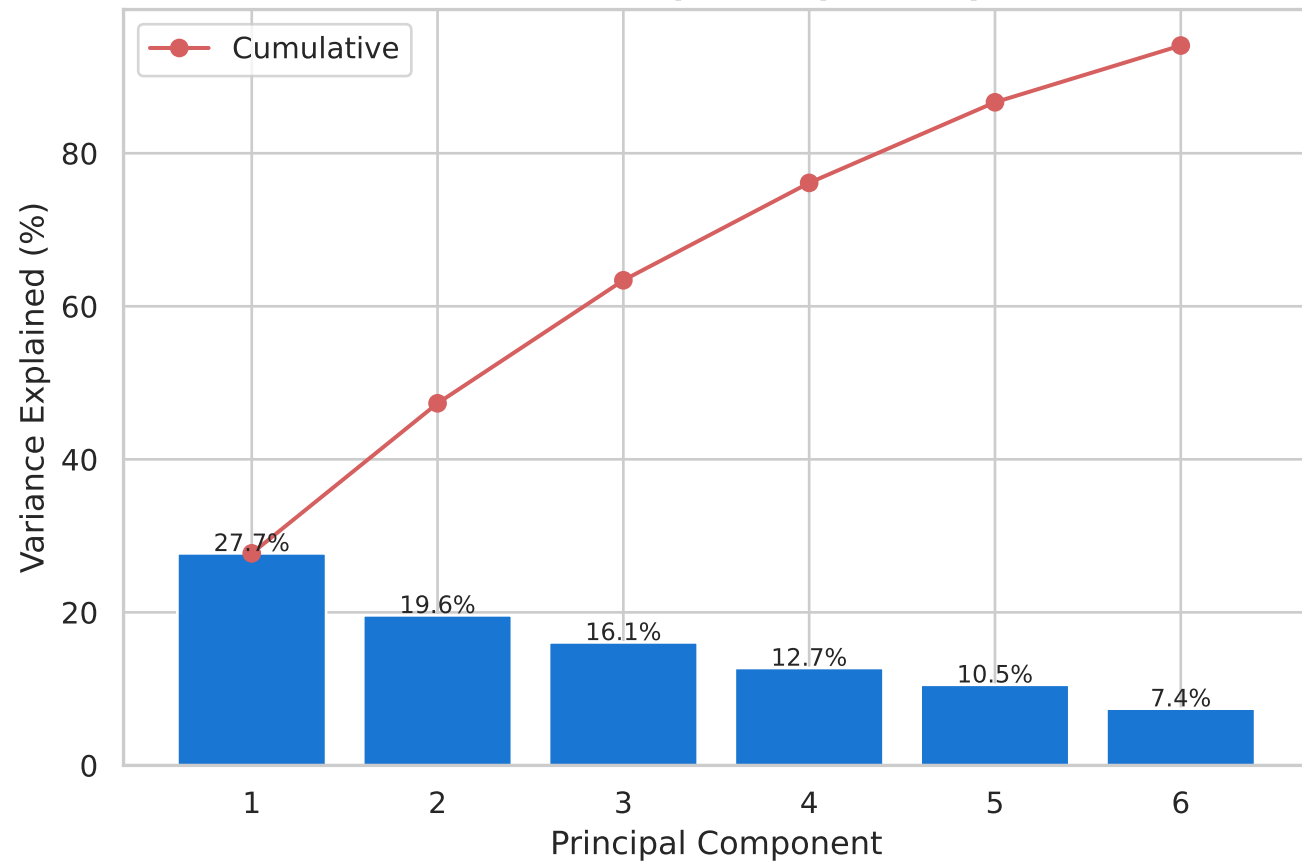
Availability Rate — Standard vs Non-standard Ports



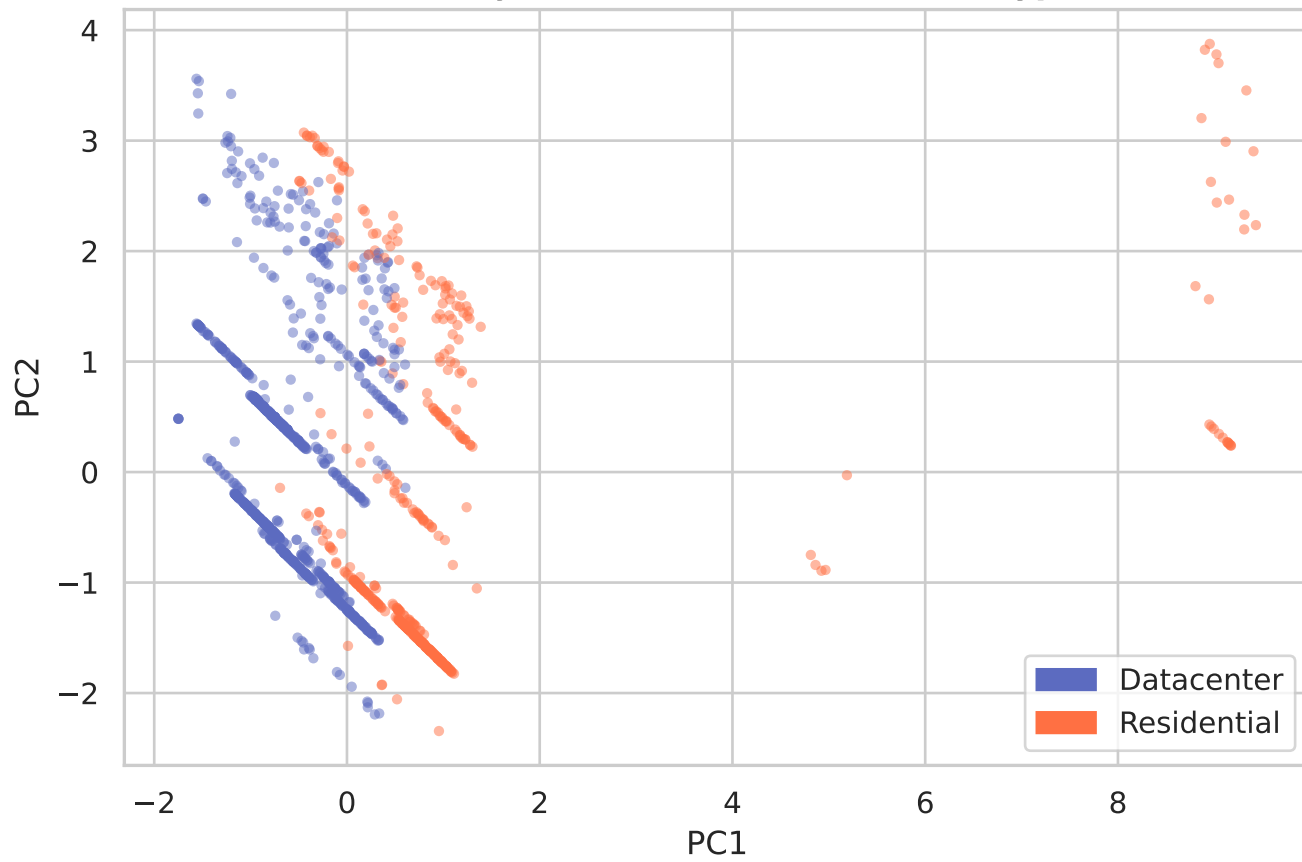
Top 15 Ports by Availability Rate
(ports with ≥ 50 configs)



PCA — Variance Explained per Component



PCA Projection (PC1 vs PC2) — Exit Type

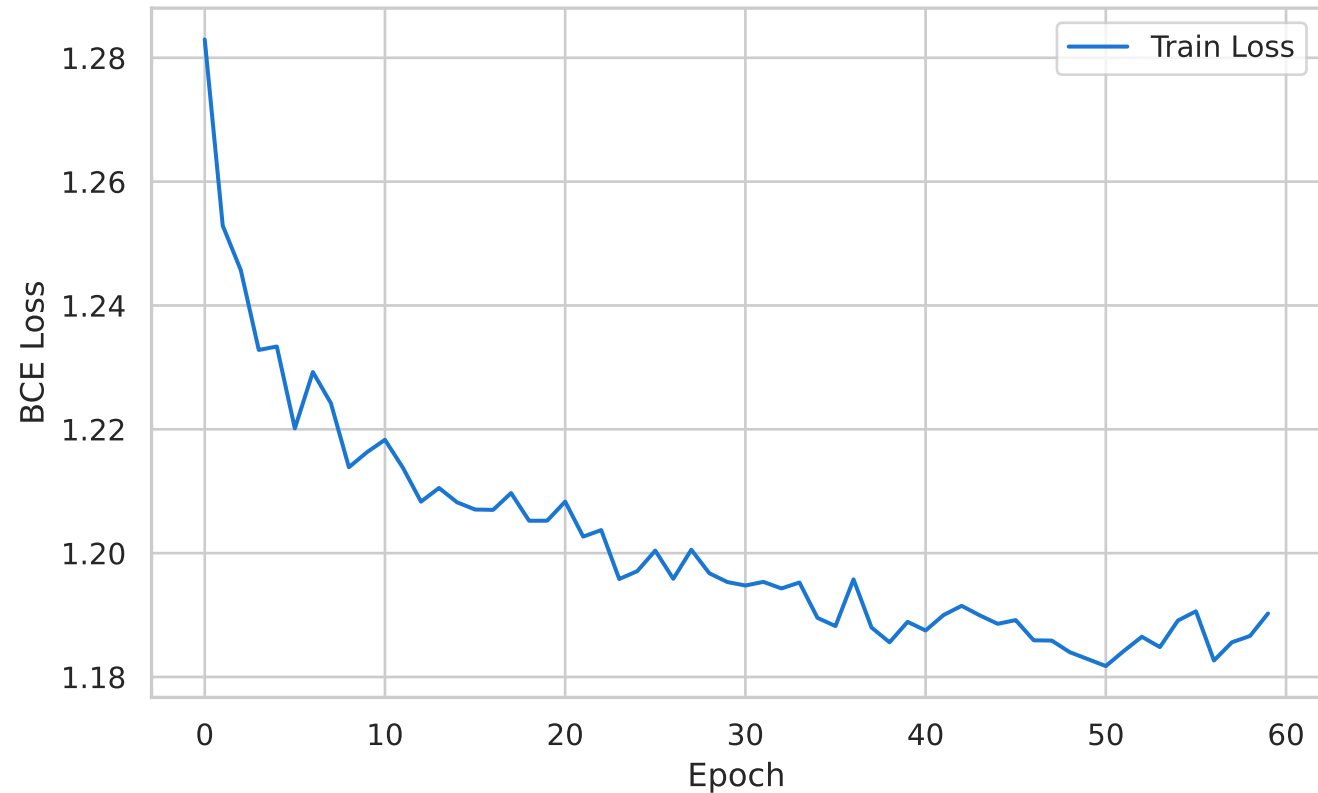


14. PCA Feature Loadings — PC1 and PC2

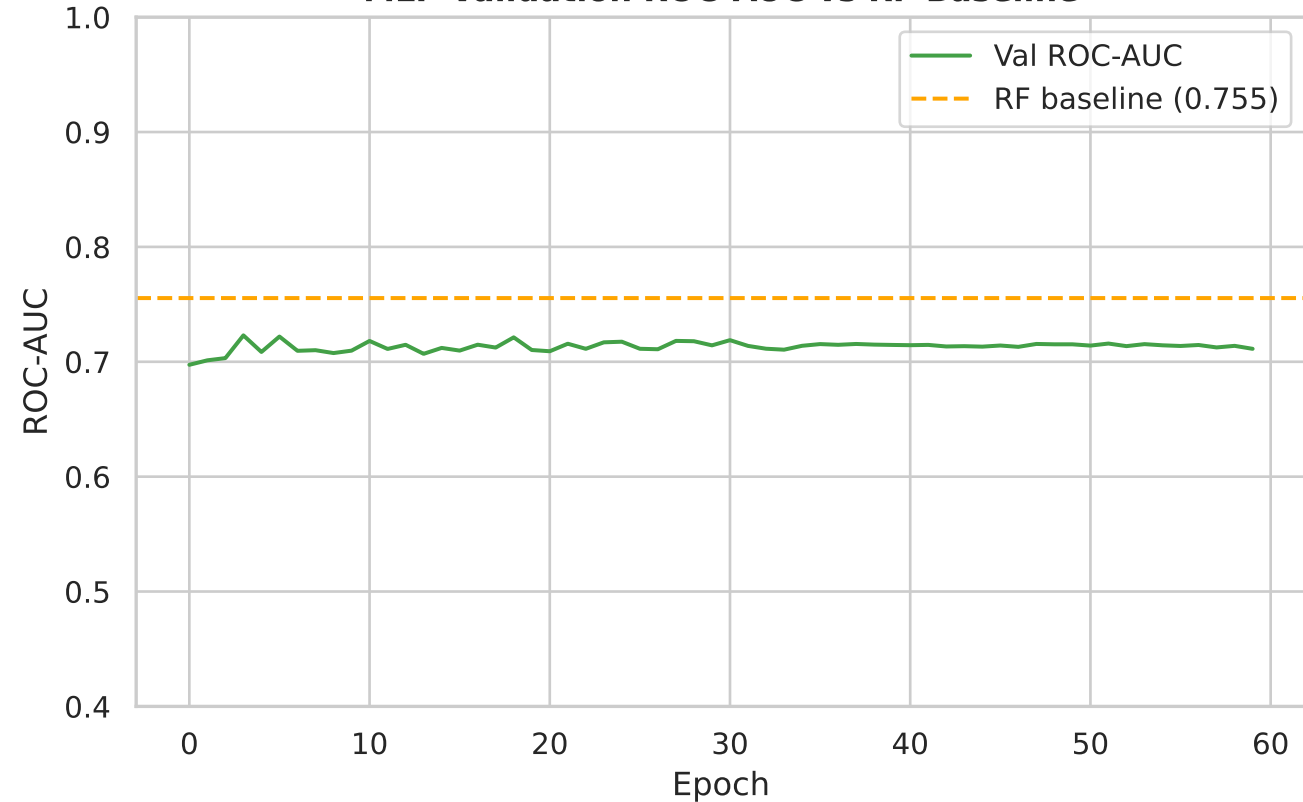
	PC1	PC2
asn_freq	0.582153	0.134568
country_freq	0.575427	0.156866
protocol_enc	0.203395	-0.242077
latency_log	0.153878	-0.131530
port_log	-0.043319	0.674882
is_standard_port	-0.125333	-0.577210
is_blacklisted	-0.341598	0.272504
is_datacenter	-0.361469	0.135969

MLP Availability Classifier — Training Curves

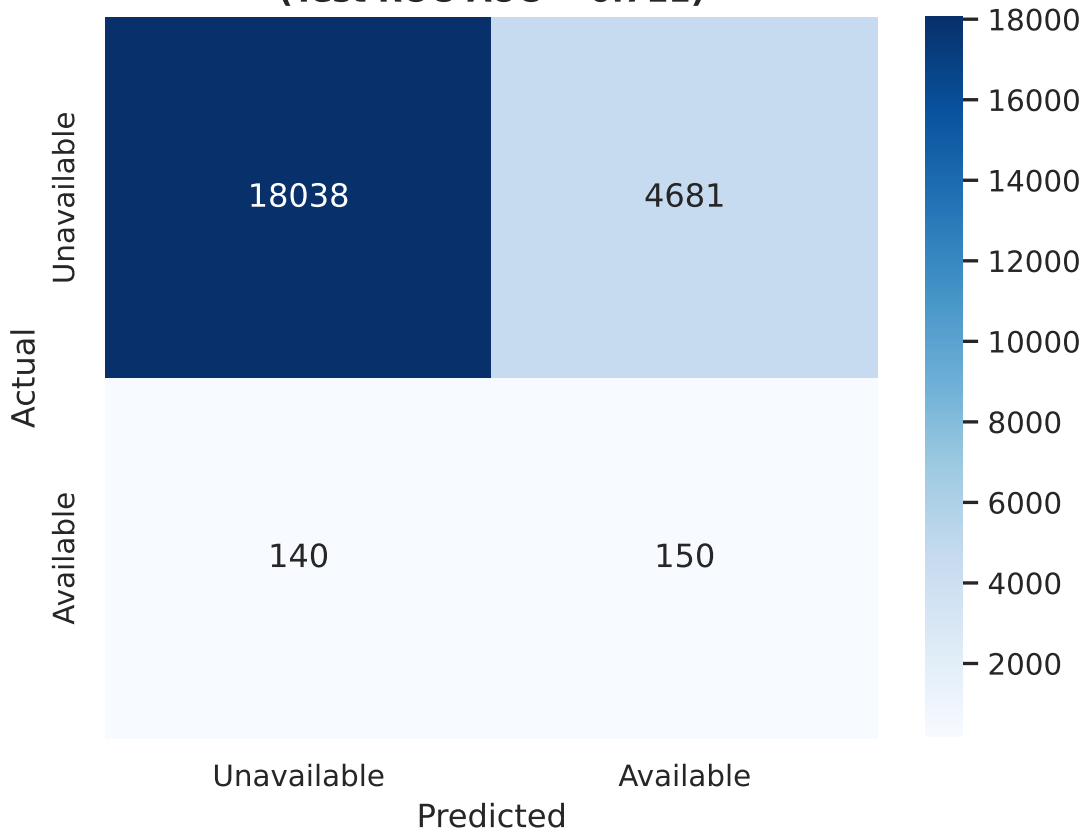
MLP Training Loss



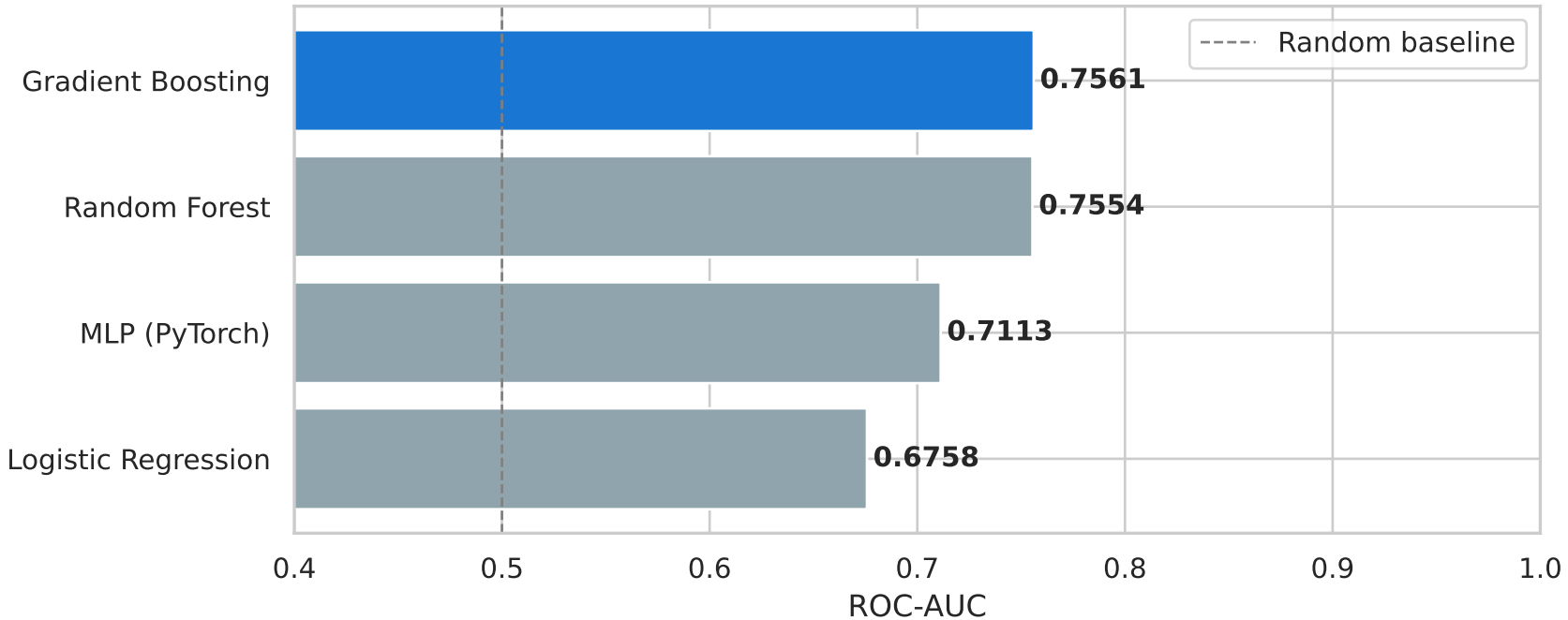
MLP Validation ROC-AUC vs RF Baseline



Confusion Matrix — MLP Availability Classifier
(Test ROC-AUC = 0.711)



Availability Prediction — Model Comparison (ROC-AUC)



17. Final Model Ranking — Availability Prediction

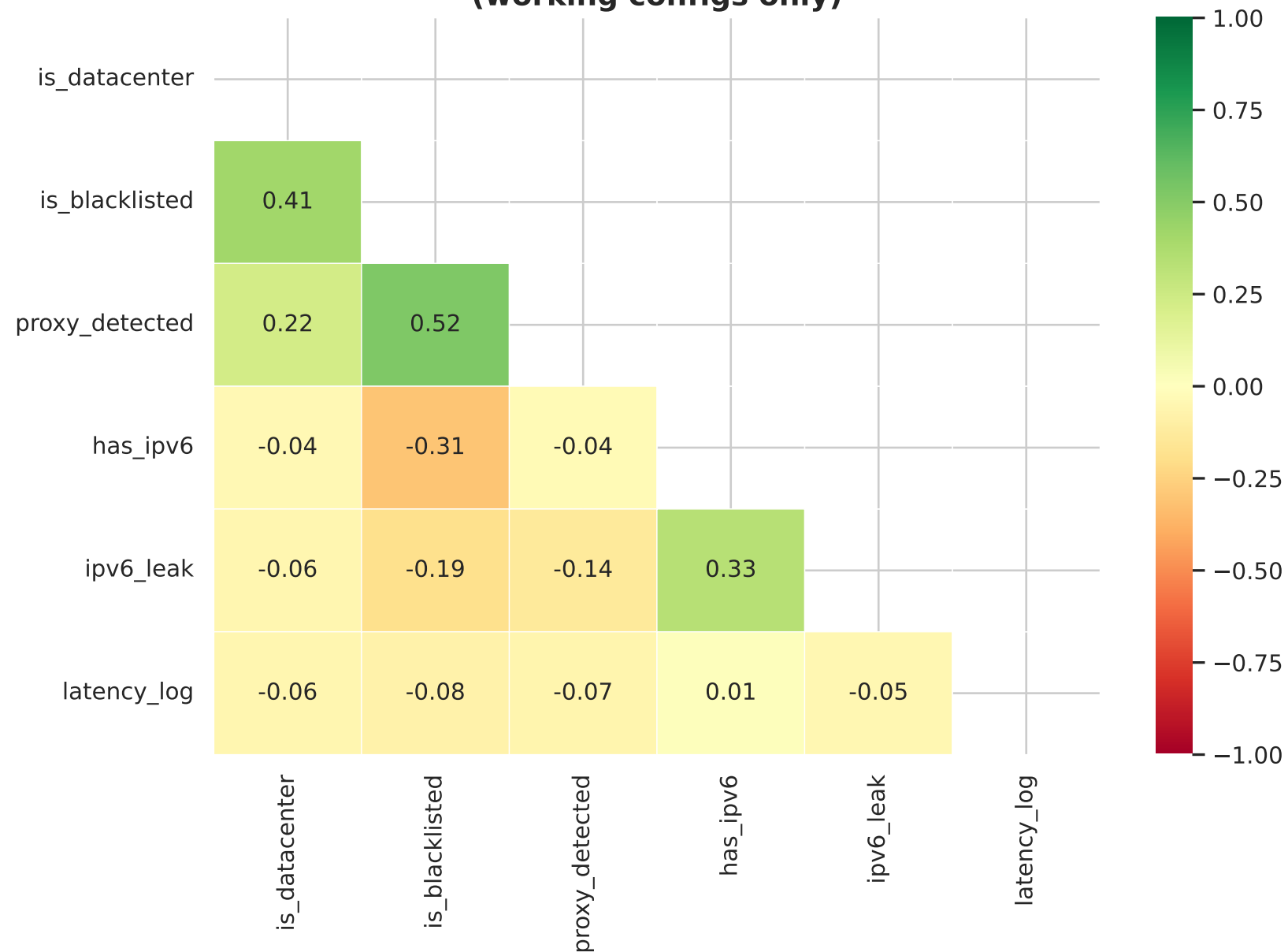
Final Model Ranking — Availability Prediction (by ROC-AUC):

1. Gradient Boosting	0.7561
2. Random Forest	0.7554
3. MLP (PyTorch)	0.7113
4. Logistic Regression	0.6758

Note: ROC-AUC > 0.5 = better than random guessing.

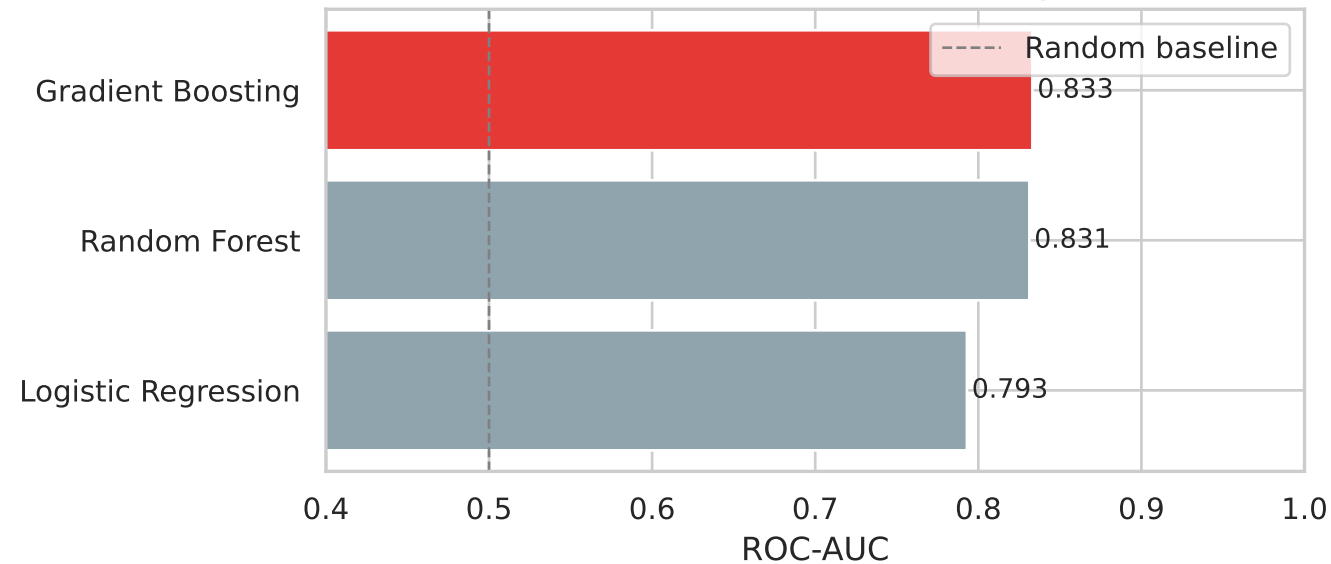
Static config metadata (protocol + port) has limited but non-trivial predictive power for availability.

Security Feature Correlations (working configs only)

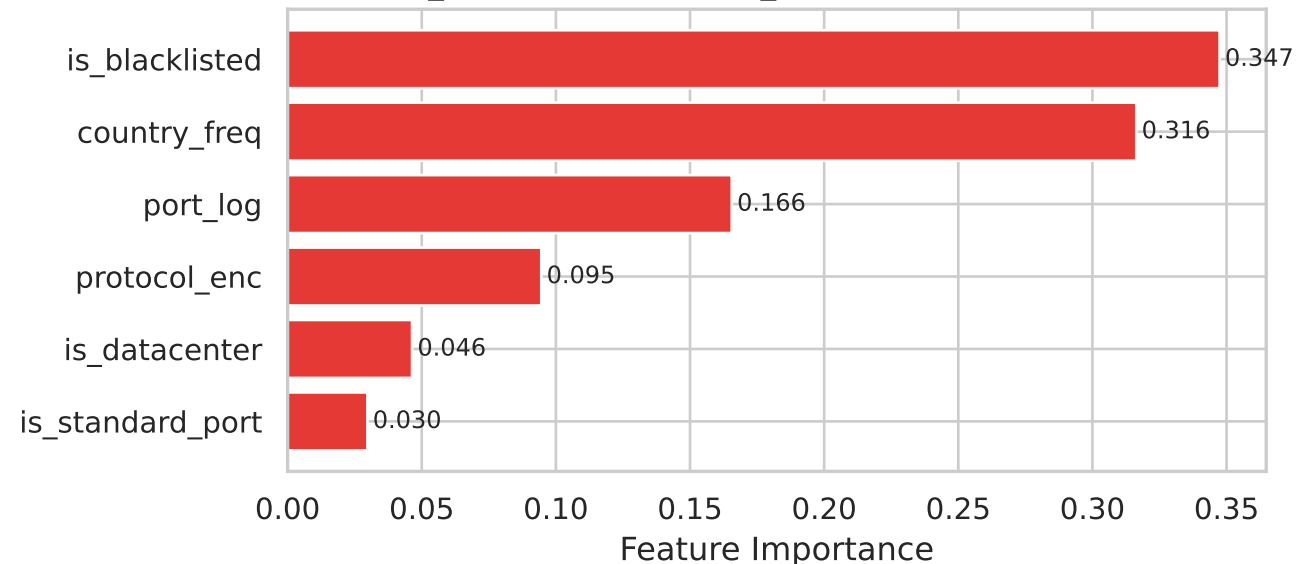


Predicting Whether a Config Will Be Flagged as a Proxy

Predicting Proxy Detection (can we know before testing?)

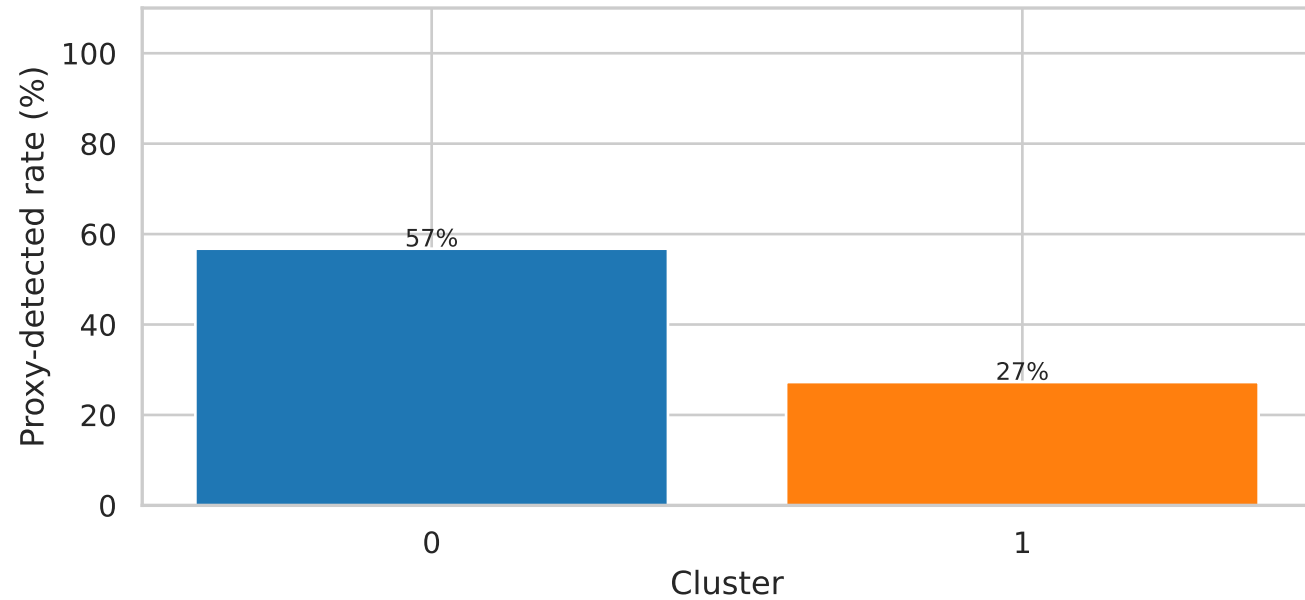


Proxy Detection — Feature Importance (is_blacklisted and is_datacenter dominate)



Security Profile by K-Means Cluster

Proxy Detection Rate per Cluster



IPv6 Support Rate per Cluster

